

Active Directory Homelab Project Report

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Overview

This project involved deploying and configuring a Windows Server 2022 Active Directory environment within Oracle VirtualBox. The aim of the lab was to simulate a small enterprise network environment and demonstrate practical systems administration and identity management skills commonly used in IT support, infrastructure, and cybersecurity roles.

The environment included:

- Windows Server 2022 Domain Controller
- Active Directory Domain Services (AD DS)
- Organisational Unit (OU) structure
- Domain user and group administration
- Windows 11 domain-joined workstation
- Group Policy Management
- PowerShell-based Active Directory enumeration

This project was completed within a virtualised lab environment to develop hands-on experience with enterprise Windows infrastructure technologies.

Objectives

The objectives of this project were to:

- Deploy a Windows Server 2022 virtual machine
 - Install and configure Active Directory Domain Services (AD DS)
 - Promote the server to a Domain Controller
 - Create a new Active Directory forest and domain
 - Configure Organisational Units (OUs)
 - Create and manage domain users and security groups
 - Perform password resets and account administration tasks
 - Join a Windows 11 client machine to the domain
 - Verify domain objects using PowerShell
 - Access and verify Group Policy Management functionality
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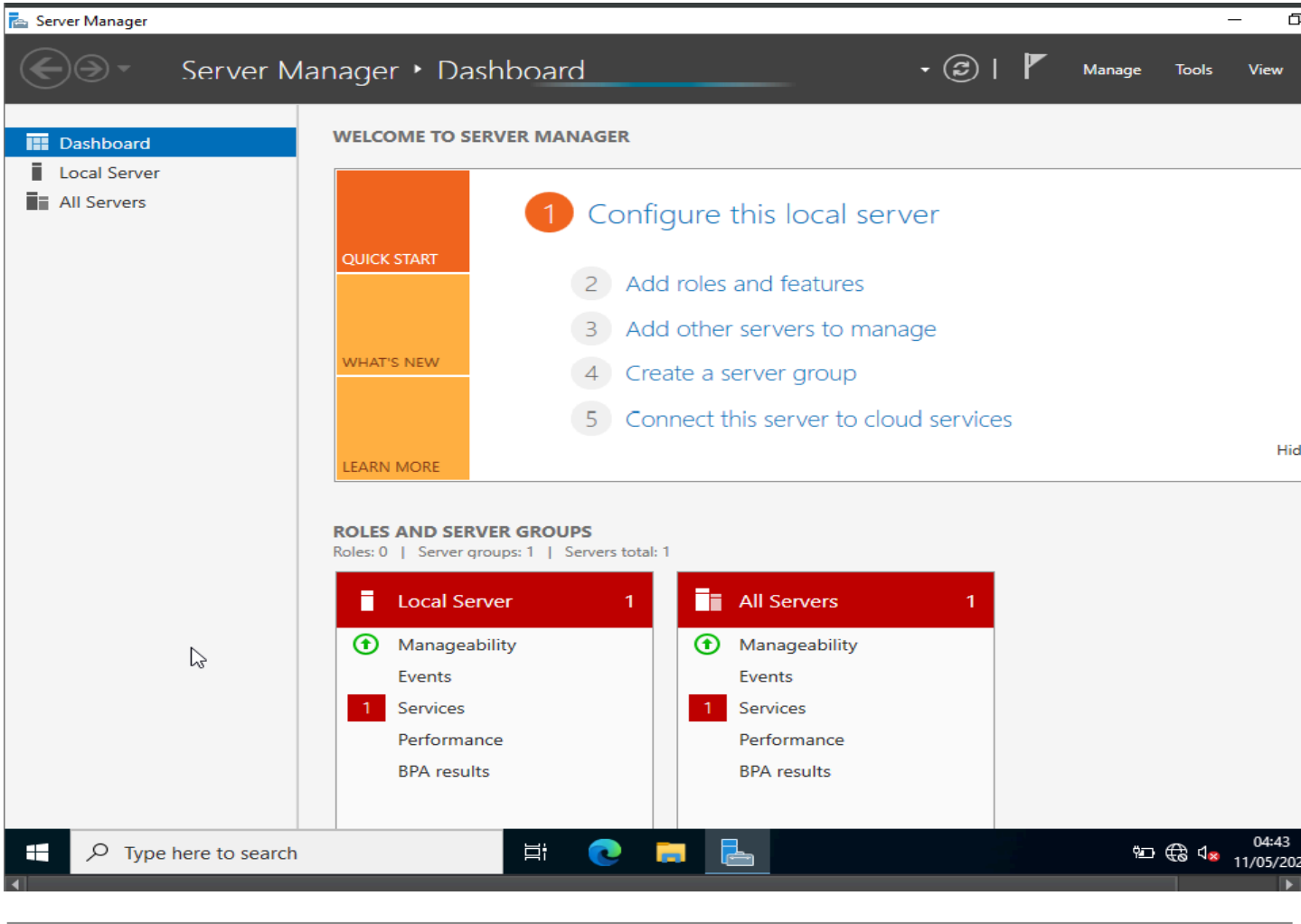
Environment Setup

The lab environment was deployed using Oracle VirtualBox.

Infrastructure Components

Component	Description
Hypervisor	Oracle VirtualBox
Server Operating System	Windows Server 2022 Standard Evaluation
Client Operating System	Windows 11
Domain Name	pavlab.local
Domain Controller	DC01

The Windows Server virtual machine was configured as the primary Domain Controller for the environment.



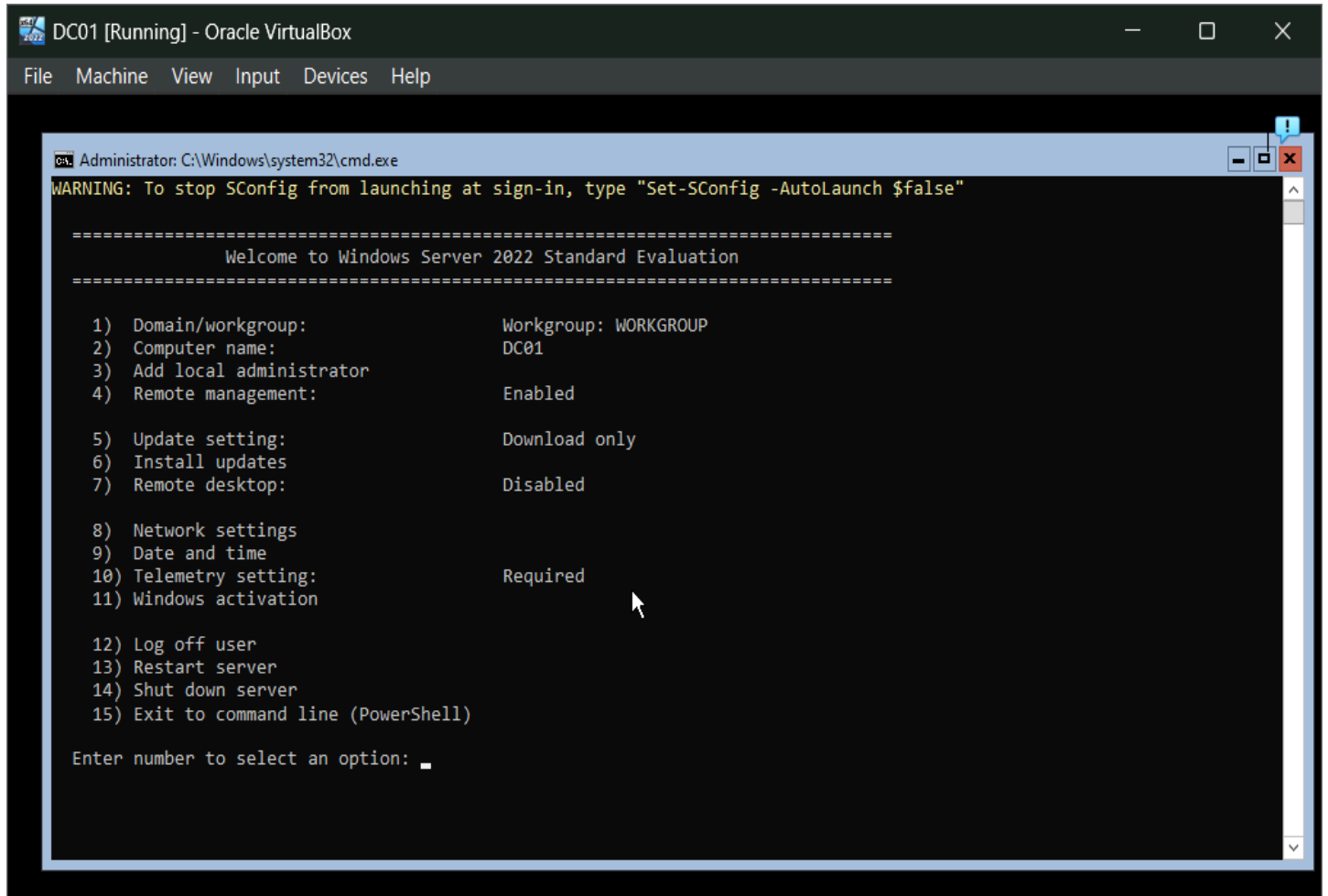
Active Directory Domain Services Installation

The Active Directory Domain Services role was installed through Server Manager.

After the installation completed, the Active Directory Domain Services Configuration Wizard was launched to begin domain controller promotion.

Tasks completed during this phase included:

- Installing the AD DS role
- Configuring the deployment operation
- Preparing the server for forest creation



Domain Controller Promotion

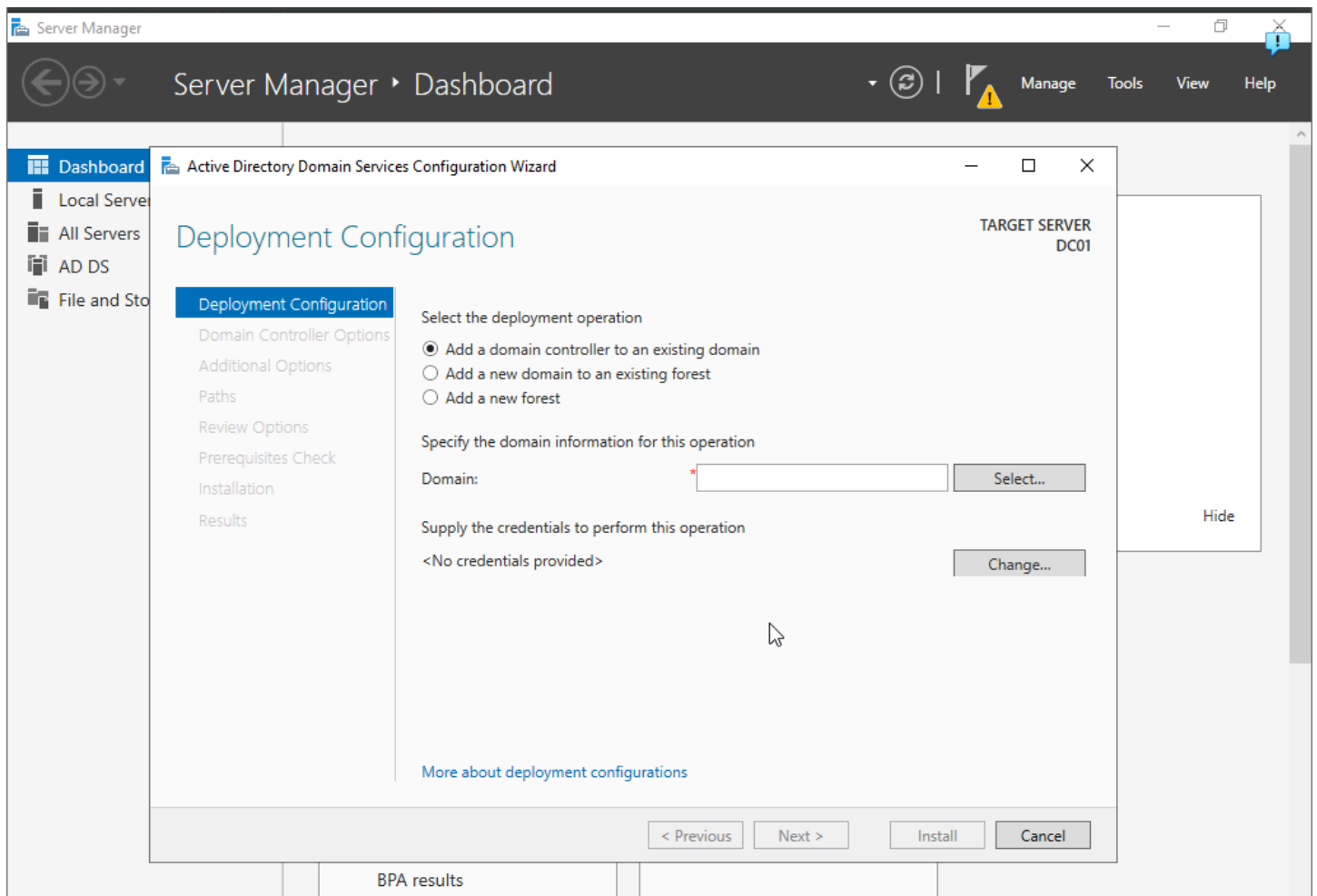
The server was promoted to a Domain Controller by creating a new Active Directory forest.

Domain Configuration

Setting	Value
Domain Name	pavlab.local
NetBIOS Name	PAVLAB
Server Name	DC01

The domain controller promotion process configured the server to provide:

- Authentication services
- Directory services
- Centralised identity management
- Domain infrastructure services



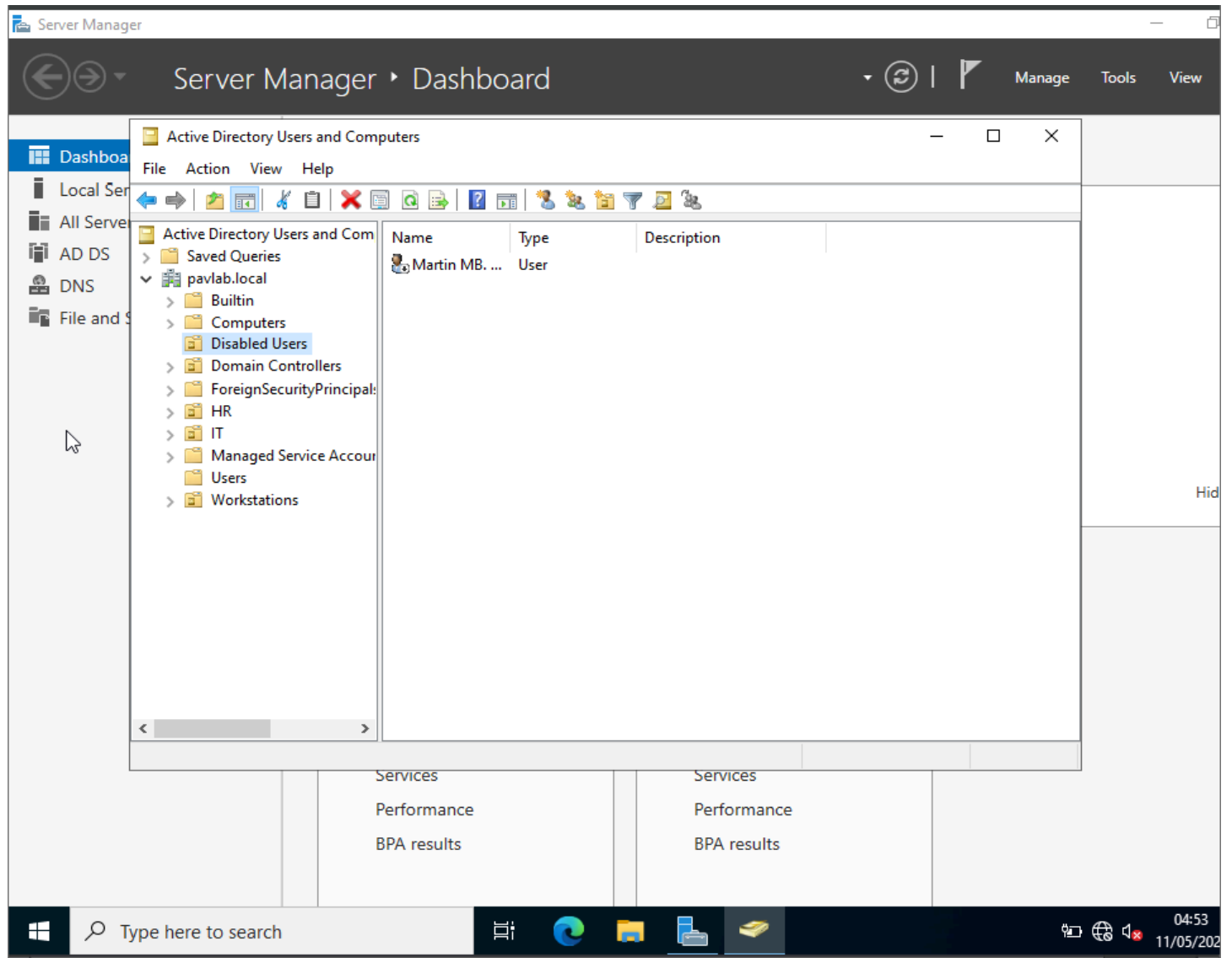
Organisational Unit (OU) Structure

A structured Organisational Unit hierarchy was created within Active Directory Users and Computers (ADUC).

Organisational Units Created

- HR
- IT
- Workstations
- Disabled Users

The OU structure allows administrative separation and enables future Group Policy targeting.



User Account Administration

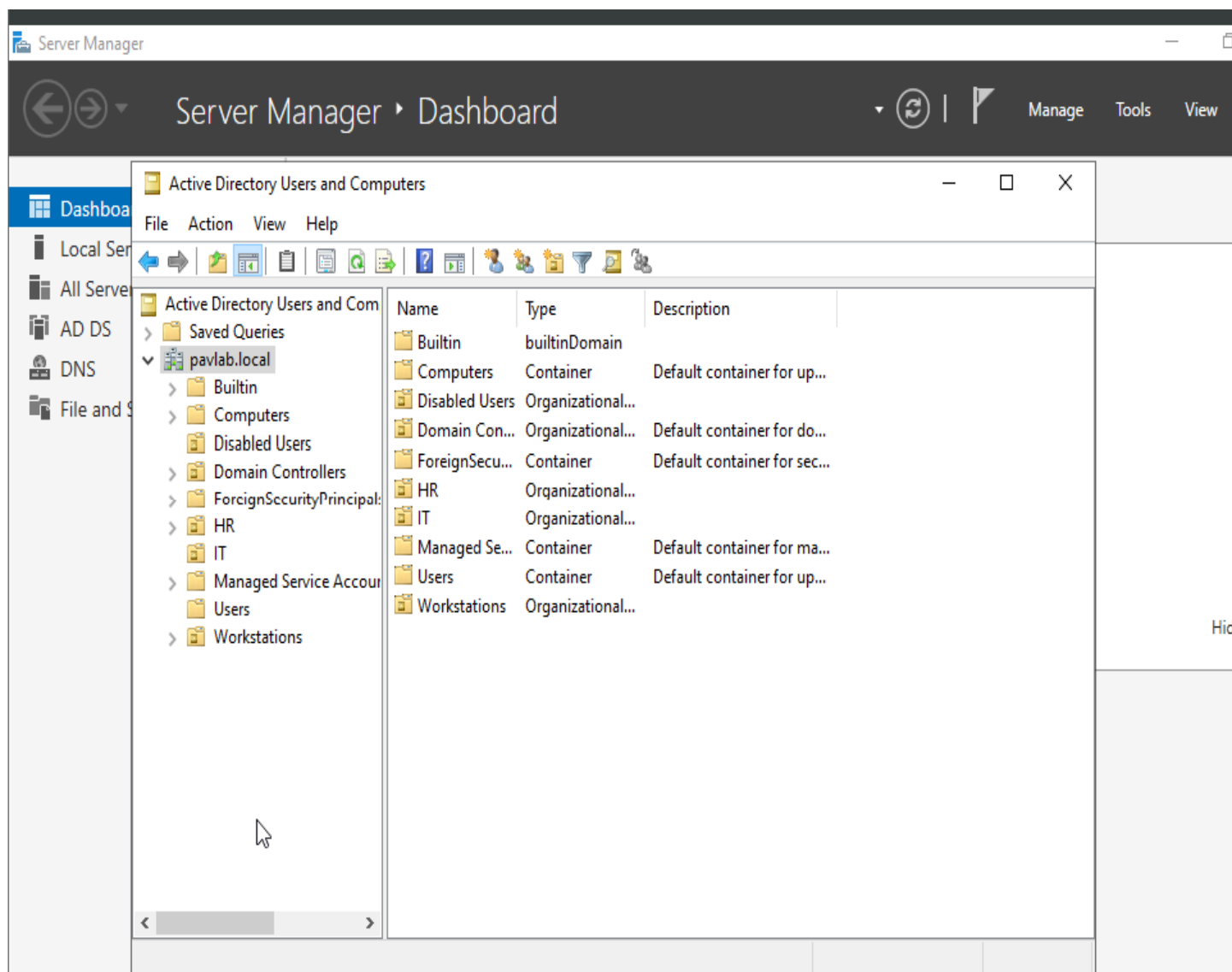
Several domain user accounts were created and configured within Active Directory.

User Accounts Created

Name	Username	Status
John JS. Smith	jsmith	Enabled
Martin MB. Brown	mbrown	Disabled

Configuration tasks included:

- Creating user logon names
- Configuring User Principal Names (UPN)
- Managing password settings
- Reviewing account properties
- Disabling accounts



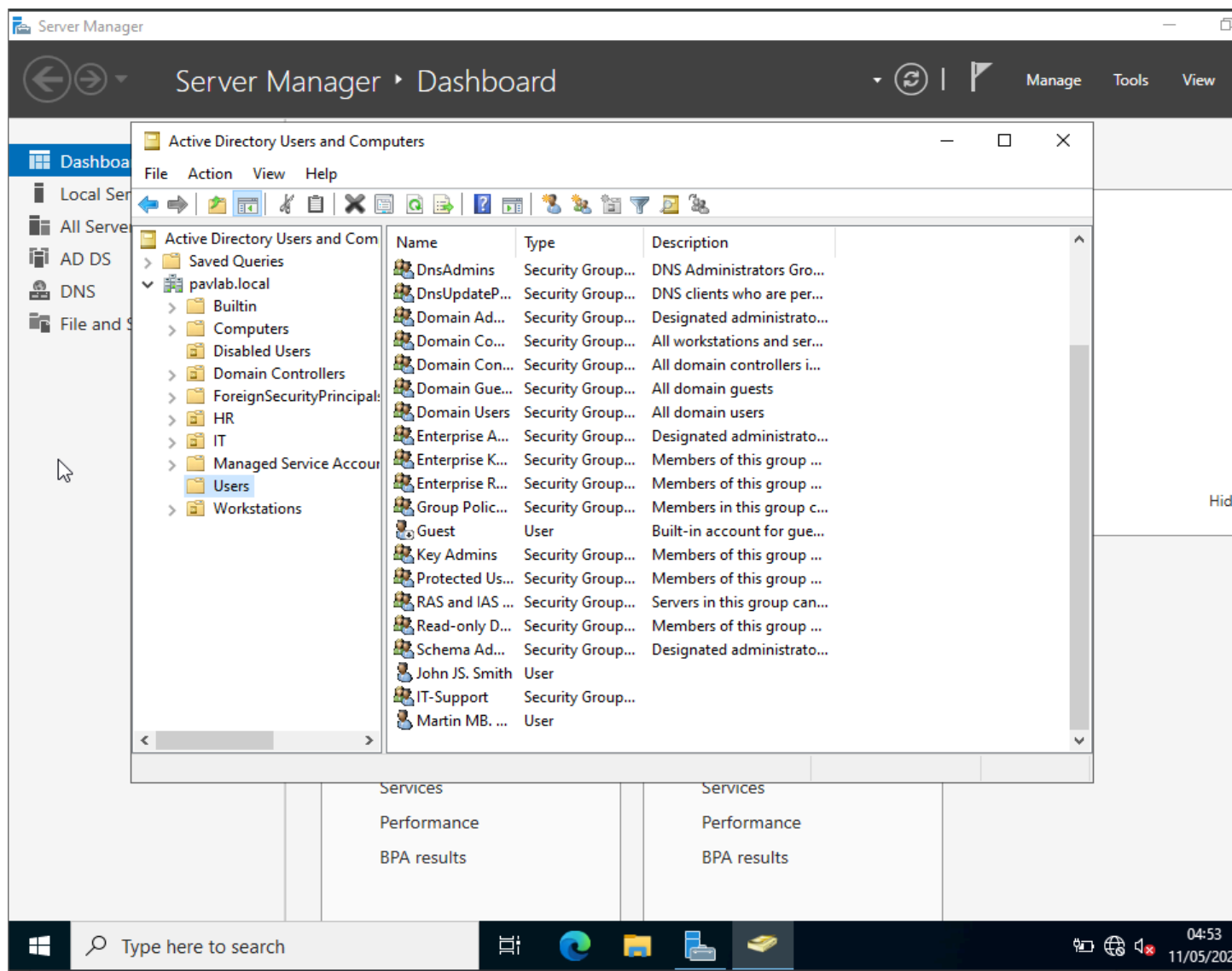
Security Group Management

A security group named **IT-Support** was created to demonstrate role-based access control.

Tasks Performed

- Created the IT-Support security group
- Added users into the group
- Verified group membership
- Reviewed group member relationships

This demonstrates how Active Directory security groups simplify permissions management within enterprise environments.



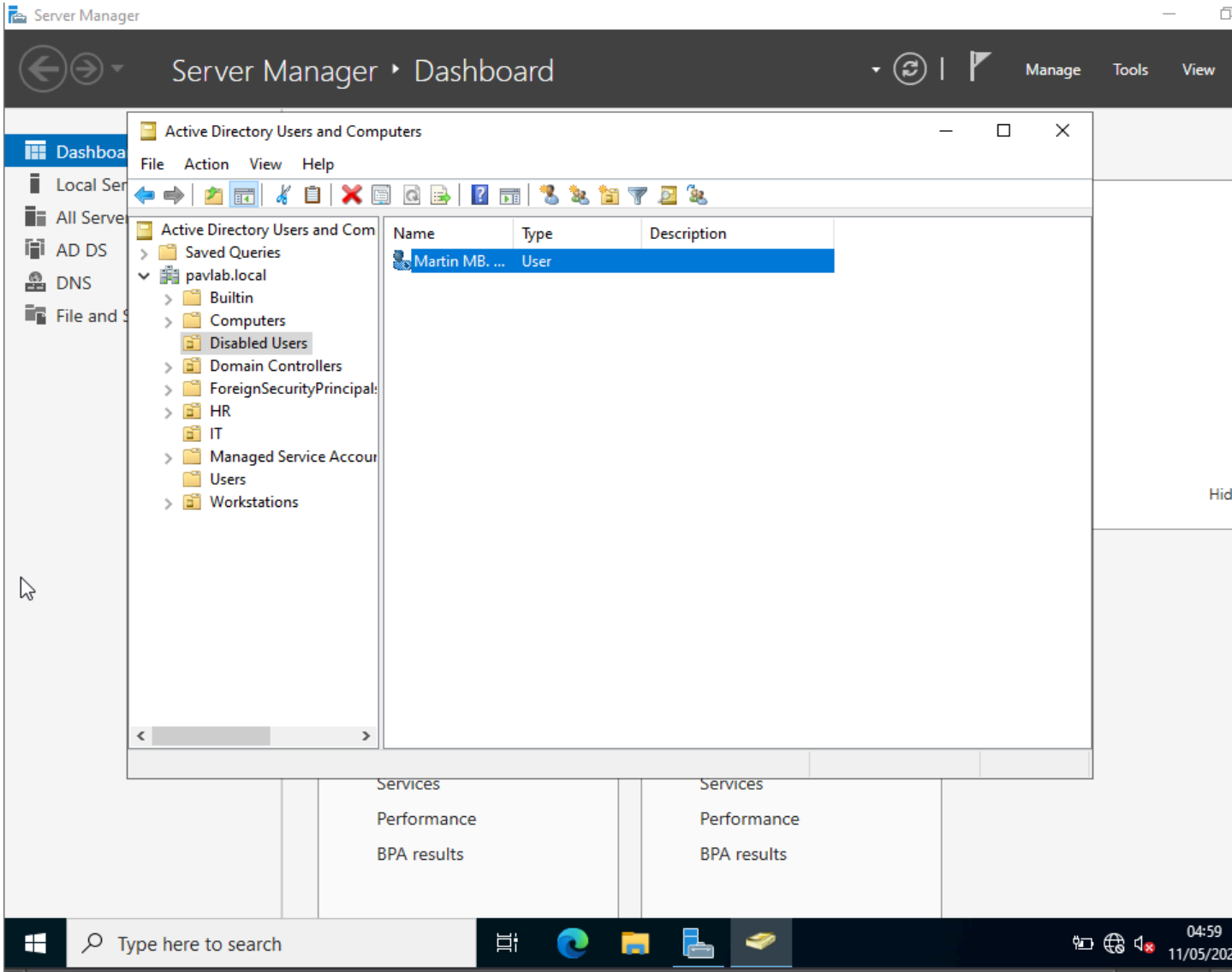
Password Reset and Account Life-cycle Management

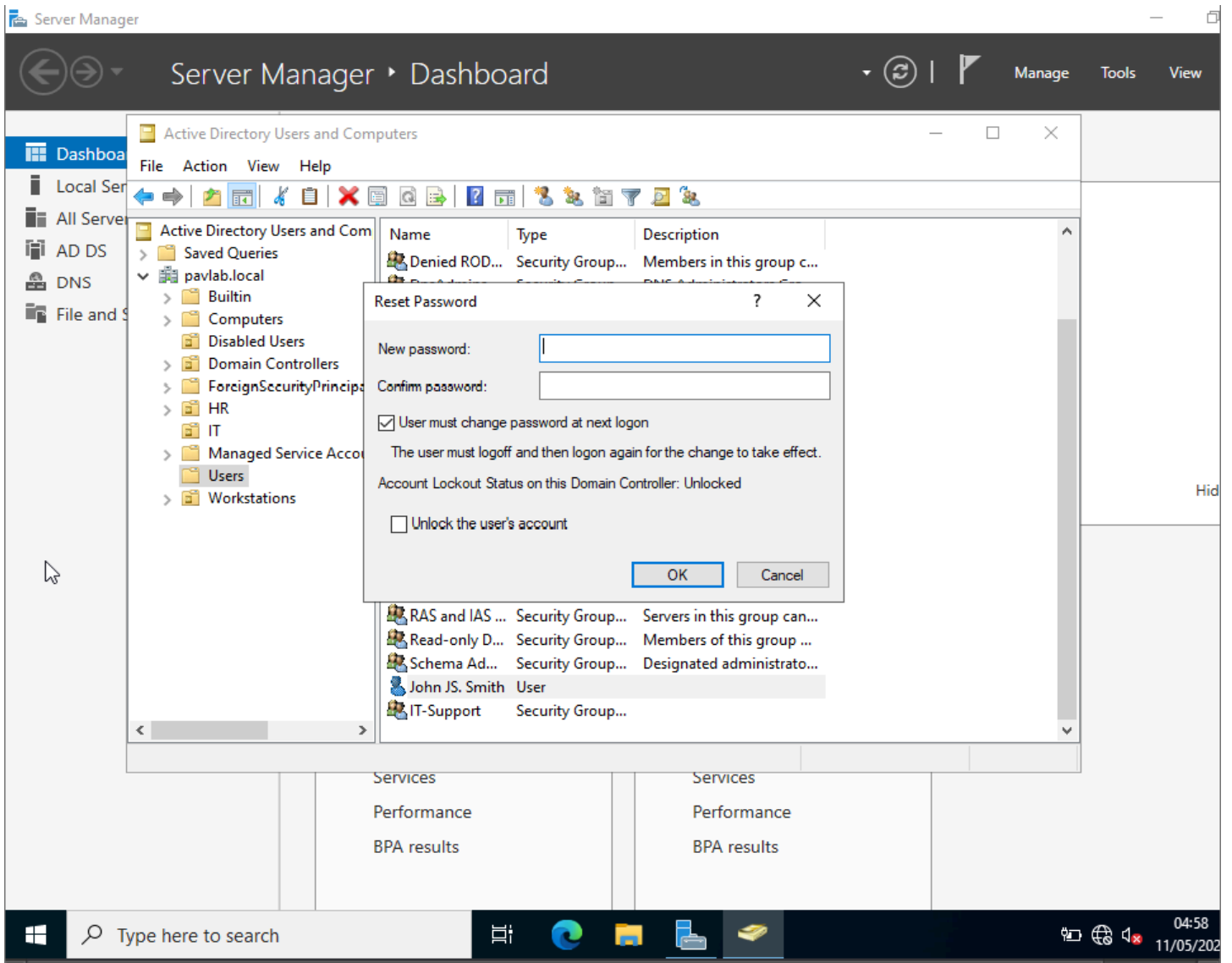
Administrative account management procedures were performed using Active Directory Users and Computers.

Tasks Completed

- Reset user passwords
- Forced password change at next logon
- Disabled user accounts
- Verified disabled account status

These are common administrative tasks performed by IT support and systems administration teams.





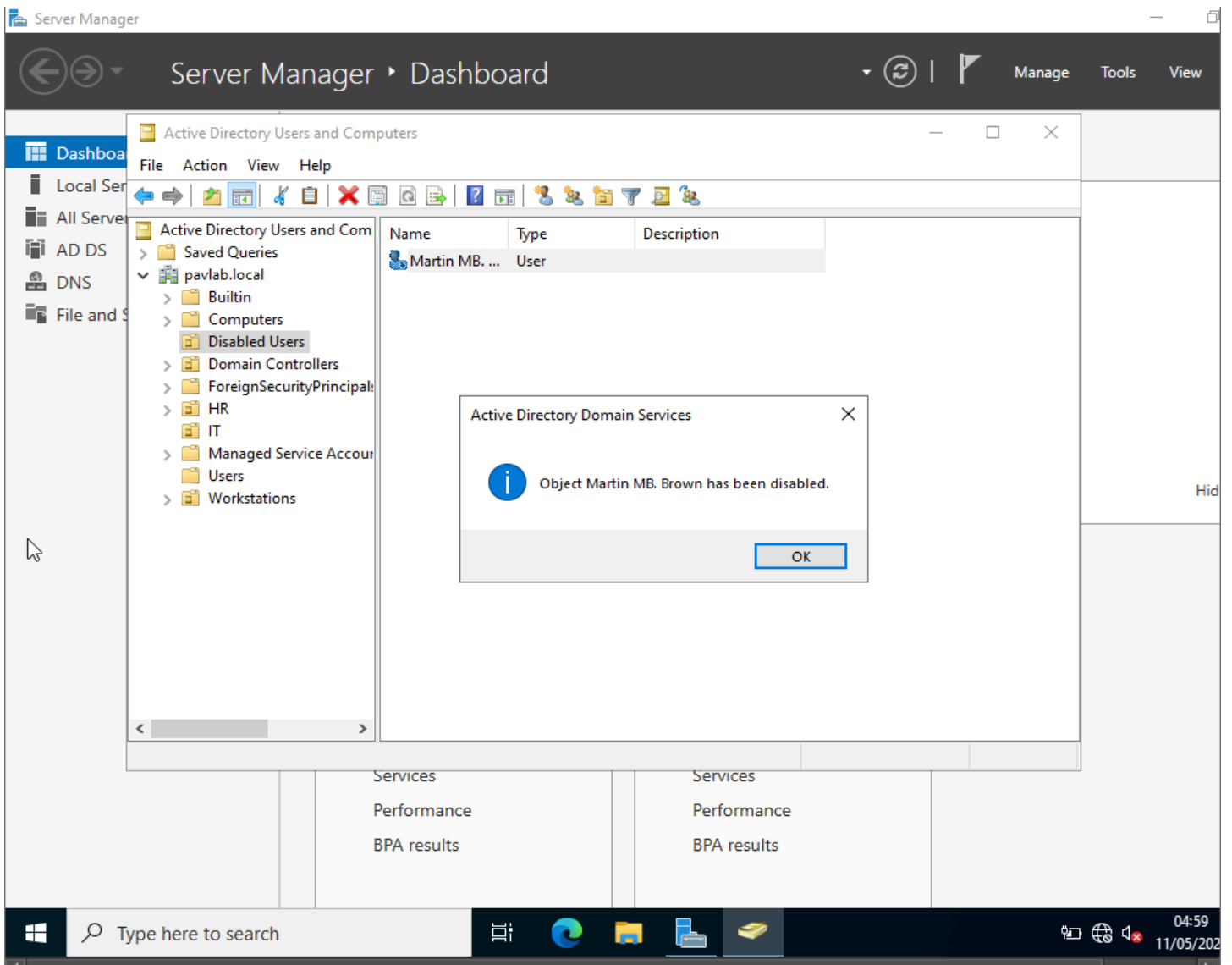
Windows 11 Workstation Domain Join

A Windows 11 client machine was joined to the `pavlab.local` domain.

After joining the domain:

- The workstation object appeared inside Active Directory
- The device was verified inside the Workstations OU
- Domain communication was successfully established

This process demonstrates centralised workstation management within a domain environment.

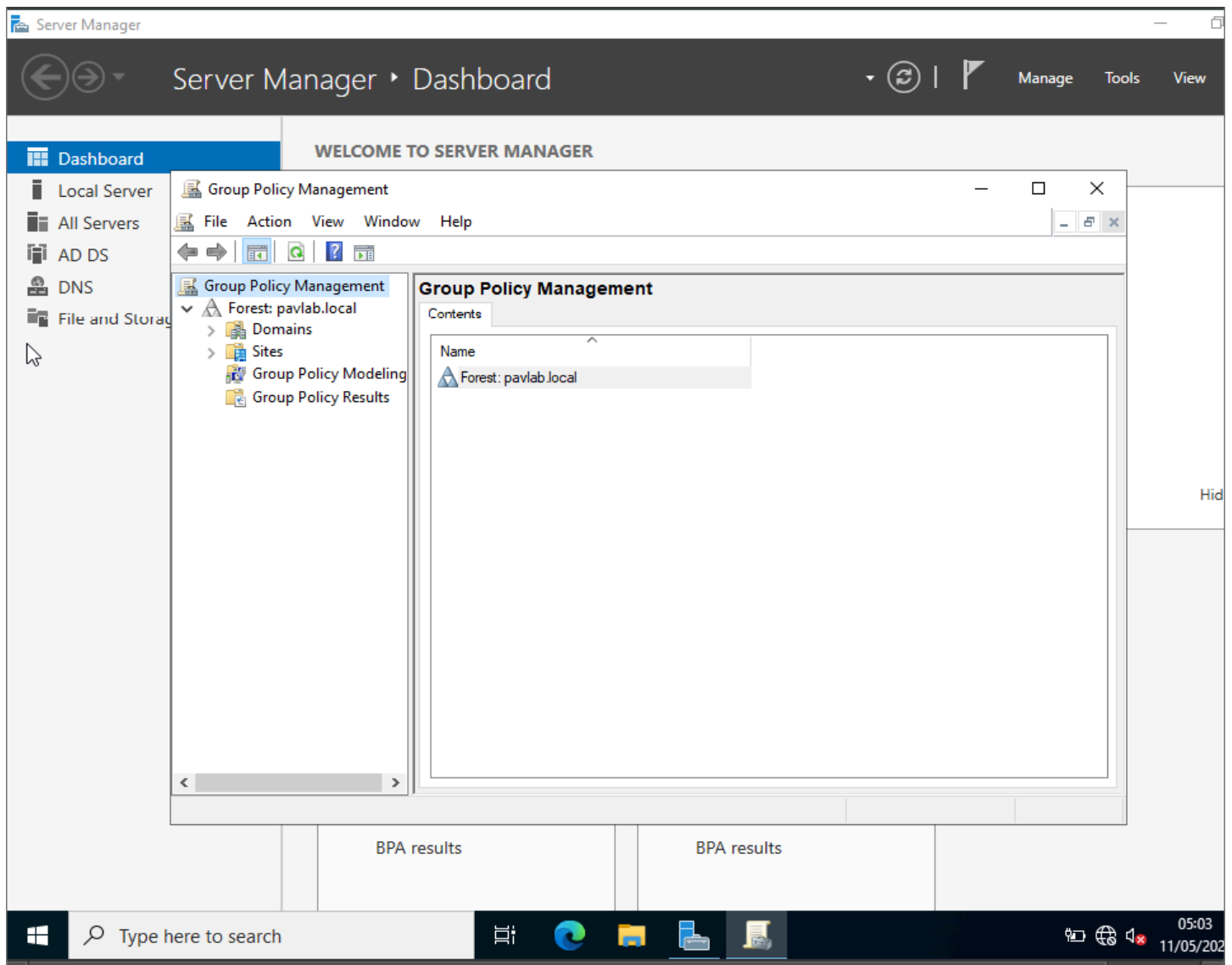


Group Policy Management

The Group Policy Management Console (GPMC) was accessed and verified following Active Directory deployment.

The Group Policy infrastructure allows administrators to:

- Enforce security policies
- Configure workstation settings
- Apply centralised administrative controls
- Standardise enterprise configurations



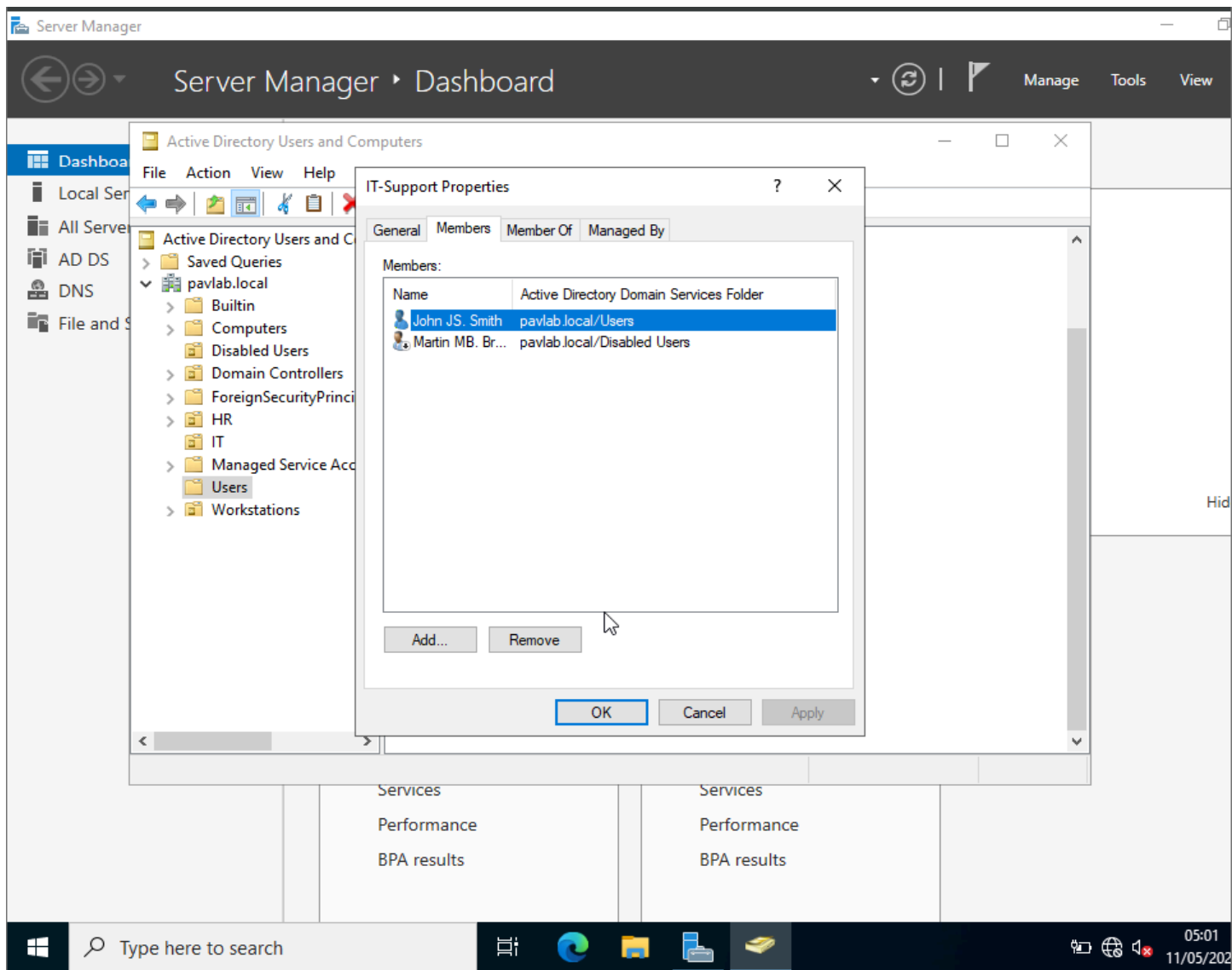
PowerShell Enumeration and Verification

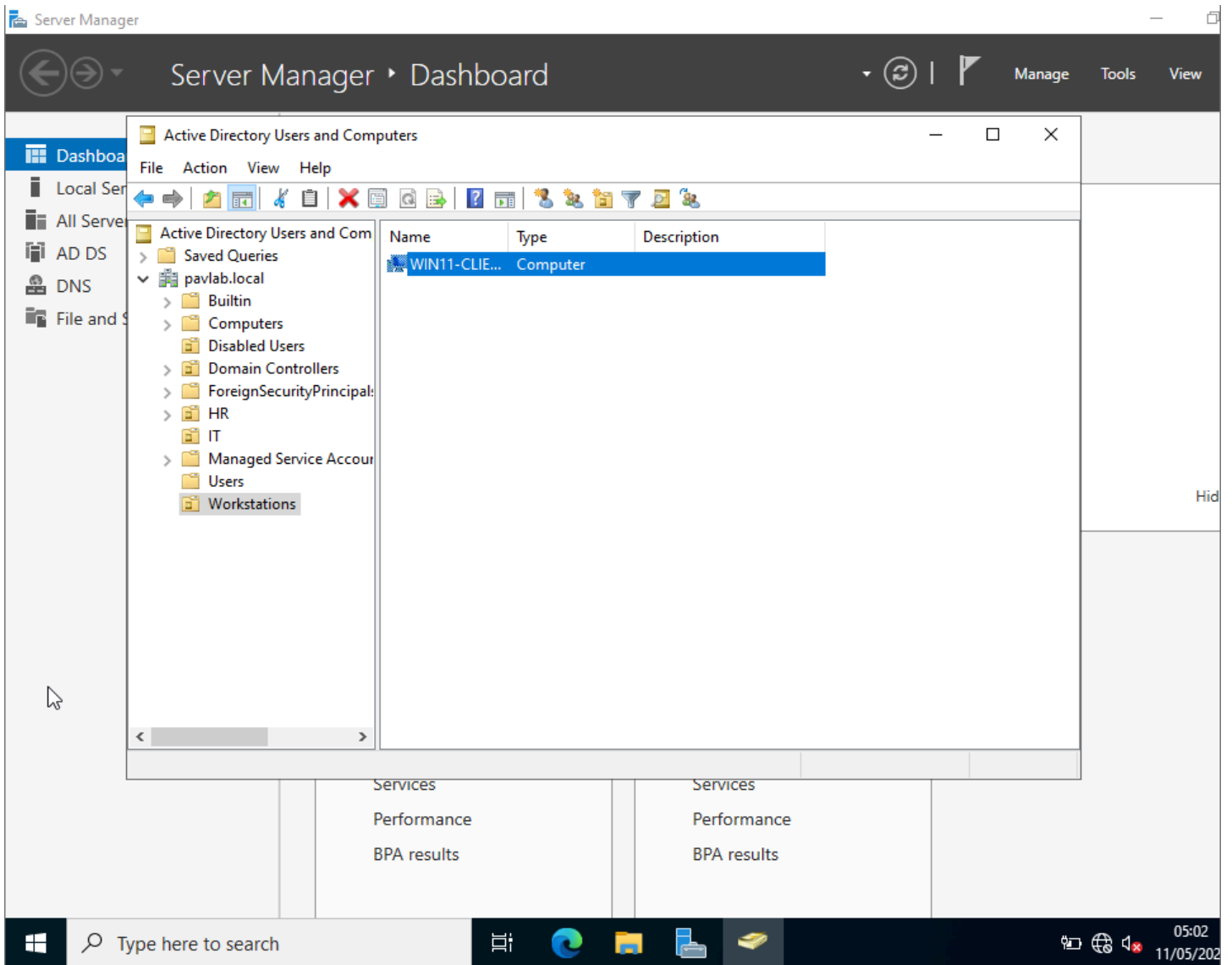
PowerShell was used to enumerate and verify Active Directory objects and domain information.

Administrative Tasks Performed

- Retrieved Active Directory domain details
- Verified domain configuration information
- Enumerated user account information
- Reviewed enabled and disabled account states

PowerShell provides scalable administration and automation capabilities within enterprise Windows environments.





Technologies Used

- Windows Server 2022
- Active Directory Domain Services (AD DS)
- Active Directory Users and Computers (ADUC)
- Group Policy Management Console (GPMC)
- Windows PowerShell
- Oracle VirtualBox
- Windows 11

Skills Demonstrated

This project demonstrates practical experience in:

- Active Directory administration
- Windows Server deployment
- Domain Controller configuration

- Organisational Unit management
 - User and group administration
 - Identity and Access Management (IAM)
 - Password and account life-cycle management
 - Group Policy Management
 - Windows workstation domain integration
 - PowerShell administration
 - Virtualised lab deployment
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Conclusion

This project successfully demonstrated the deployment and administration of a Windows Active Directory environment using Windows Server 2022 and Oracle VirtualBox.

Through this lab, practical experience was gained in:

- Enterprise identity management
- User and group administration
- Domain infrastructure deployment
- Centralised workstation management
- Active Directory administrative tools
- PowerShell-based systems administration

The project provided hands-on exposure to technologies and administrative tasks commonly performed within IT support, infrastructure, and cybersecurity environments.